Blessed Madukoma

Software & Research Engineer | LLM agents and AI data systems

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Summary

Software and research engineer with 3+ years building backend and data systems in Python and Go, now focused on LLM agents. Combines research and engineering to make AI-powered data systems give answers that are complete and up to date, or clear about what they lack. MSc, Carnegie Mellon University.

Experience

- **Graduate Research Associate** *Jun 2026 – Sep 2026*

  ***Carnegie Mellon University Africa**, Spatial and Language Technologies (SaLT) Lab* Kigali, Rwanda

  - Traced 37% of DataAgentBench's recorded agent failures (16,293 of 44,277) to its harness, not its six LLMs (including GPT-5.2 and Gemini 3 Pro); removing them reorders the models by failed steps. Analysed 283,435 agent steps across six benchmarks.
  - Built a static guardrail in Python (`ast`) that blocks agent code naming missing files, columns or keys before it runs, with zero false blocks on 8,517 working steps from three unseen benchmarks; on DABstep, a standard linter (pyflakes) flags only 6% of the errors it stops.
  - Built a sandboxed agent harness (no network, read-only data) and ran 36,990 pre-registered runs: for gpt-oss-120b, the guardrail cut errors that ran from 3,316 to 2 and raised accuracy by 1.44 points; no detectable gain for gemma4-31b or over a cheaper error message listing valid names.

- **Backend Software Engineer (Contract)** *Jun 2026 – Aug 2026*

  SLS Energy Kigali, Rwanda

  - Cut invoicing from over a month after month-end to two days by replacing a manual spreadsheet with a metered billing engine (Python, FastAPI, PostgreSQL) that now invoices 9 client organisations.
  - Ensured every invoice reflects measured energy only, by excluding and flagging days with missing meter readings instead of estimating them.
  - Closed all 18 critical and high-severity findings from two security audits, including 8 cross-tenant data-exposure flaws; set up CI gates (pytest at 80% coverage, mypy, ruff, pip-audit, gitleaks).

- **Graduate Research Assistant** *Sep 2025 – May 2026*

  ***Carnegie Mellon University Africa**, Spatial and Language Technologies (SaLT) Lab* Kigali, Rwanda

  - Narrowed 31,604 athletes to about 163 for investigator review (99.5% fewer) by combining 8 statistical, ML and Bayesian detectors over 1.6M performances, checked against Athletics Integrity Unit sanctions; the paper received an IEEE SDS 2026 Best Paper Honorable Mention.
  - Replaced a Kafka-based, four-container pipeline with one FastAPI service on PostgreSQL, DuckDB and Redis: a full analysis fell from 129 s to 33 s, and testing a method change from 20+ min to about 30 s.

- **Data Engineering & Analytics Engineer (Contract)** *May 2025 – Sep 2025*

  Ministry of Infrastructure (MININFRA) Kigali, Rwanda

  - Architected Rwanda's infrastructure intelligence platform, built to attract foreign investment: pipelines that join construction-permit and land-use records, and a Grafana dashboard of 12+ indicators such as permit throughput and capital-allocation trends.

- **Software Engineer** *Jun 2025 – Sep 2025*

  Huza HR Kigali, Rwanda

  - Built the email-notification system (candidate, recruiter and internal templates with delivery tracking) behind the company's first fully digital applicant-tracking workflow, in Next.js and Node.js.

- **Software Engineer** *Sep 2024 – Dec 2024*

  The UPANZI Network Kigali, Rwanda

  - Cut page load times by 40% on an African dataset-hosting platform by keeping fetched data in Redux state, which cut repeated database queries; cut the CSS bundle by 30% with Tailwind CSS.

- **Golang Backend Engineer (Contract)** *Mar 2024 – Aug 2024*

  Platnova Remote

◦ Cut payment processing time by 20% by automating vault settlement in Go with scheduled jobs, and support enquiries by 15% by redesigning transaction receipts.

- **Software Architect (National Youth Service)** *Dec 2022 – Nov 2023*

21st Century Technologies Lagos, Nigeria

◦ Cut release time by 40% by automating deployments (CI/CD with Docker and GitHub webhooks), and co-led an internal reporting system that employees used to give feedback on deployed projects.

◦ Coordinated DNS service restoration and implemented failover protocols across Tier III/IV data centres; uptime held at 99.99%.

Projects

- **Sedge** [**sedge.app**]

Tools: Go, PostgreSQL, MySQL, Redis

◦ Built the Go backend of a data platform that answers questions over company databases with LLM-written SQL (text-to-SQL).

◦ Wrote its query guardrail, now in code review: one read-only SELECT with allow-listed functions, a 15-second timeout and a 10,000-row cap; it blocks all 61 attack queries written to test it.

- **data-agent-traces** [**Code**, **DOI**]

Tools: Python

◦ Open-sourced the trace parsers, guardrail and experiment scripts behind the agent-trace study, archived on Zenodo.

- **Hidden-rows study** [**Report**]

Tools: Python, PostgreSQL

◦ Ran a pre-registered test of LLM data agents under PostgreSQL row-level security: on five BIRD databases, three cheaper models presented partial answers as complete in 70–95% of affected cases.

Education

- **Carnegie Mellon University** *Aug 2024 – May 2026*

MSc Information Technology (Software Engineering, Big Data & Applied ML); GPA 3.93/4.00 Kigali, Rwanda

◦ Thesis: performance anomaly detection in athletics (platform, paper). Advisor: Prof. Prasenjit Mitra.

- **Babcock University** *Aug 2018 – Jun 2022*

BSc Software Engineering, First Class Honours; GPA 4.55/5.00 Ogun, Nigeria

Skills

- **Languages:** Python, Go, SQL, TypeScript, JavaScript
- **LLM & ML:** LLM Agents, Agent Evaluation, Guardrails, Text-to-SQL, Static Analysis, Scikit-learn, Statistical Modelling
- **Backend & Data:** FastAPI, PostgreSQL, DuckDB, MySQL, Redis, Kafka, Node.js, ETL Pipelines
- **Infrastructure & Quality:** Docker, GitHub Actions (CI/CD), pytest, mypy, Git
- **Frontend:** Vue.js, Nuxt, Tailwind CSS

Selected Graduate Coursework

- **Machine Learning in Production** (Christian Kästner); **Engineering Data-Intensive Scalable Systems** (Paulo Merson): microservices, Kafka messaging, sagas, CQRS, event sourcing, circuit breakers and observability.

Publications

[1] **B. Madukoma**, P. Mitra (2026). Performance Anomaly Detection in Athletics: A Benchmarking System with Visual Analytics. *IEEE Swiss Conference on Data Science and AI (SDS 2026)*, Zurich. **Best Paper Honorable Mention**. arXiv:2604.21953.

[2] S. Wafula, **B. Madukoma** (2025). Unmasking COVID-19 Vulnerability in Nigeria: Mapping Risks Beyond Urban Hotspots. *Women in Machine Learning (WiML) Workshop at NeurIPS 2025*. arXiv:2509.05398.

Leadership

- **Chairman**, Academic Innovation Sub-Committee, Research Committee, CMU Africa (2024–2026).
- **Executive President**, Computer Club, Babcock University (2022): led a student tech community of 2,000+ members.